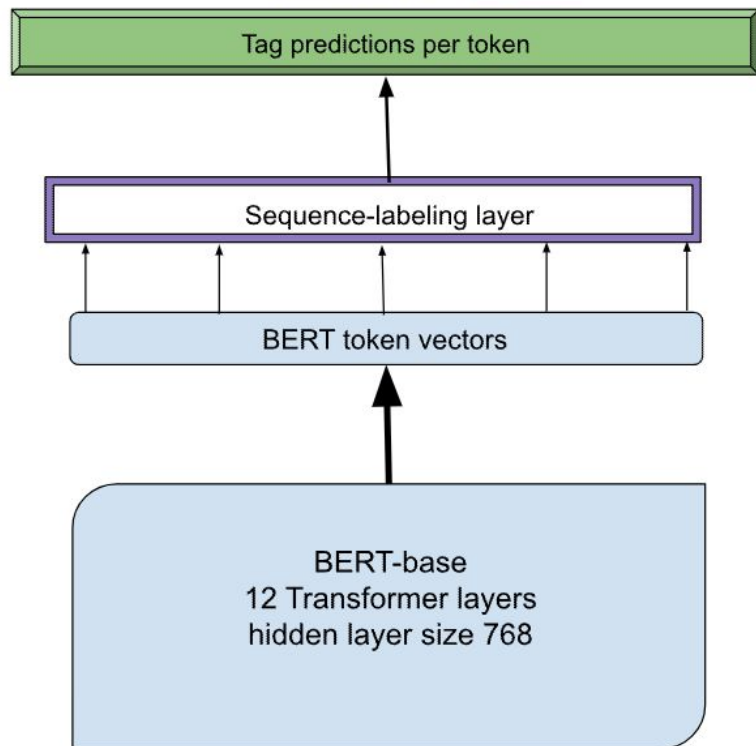


Exploring limitations of BERT for End-to-End Aspect-based Sentiment Analysis

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Model Architecture

Figure 1. Fine-tuned layer and base model



- Model outputs the **predictions per token**.
Example output ['pizza']:POS for the aspect "pizza" with a positive polarity
- Linear layer or the **sequence-labeling layer**, is used as a task specific layer to map the **BERT token vectors** to ABSA labeling, (POS, NEG, NEU).

BERT:

- Hidden layer size = 768
- Transformer layers = 12
- Feed-forward size = 3072
- Trainable parameters = 110 million

Laptop-14 Dataset

Dataset	Pos.	Neg.	Con.	Neu.	Tot.
LPT-TR	987	866	45	460	2358
LPT-TE	341	128	16	169	654

From [SemEval-2014 Task 4: Aspect Based Sentiment Analysis](#) page 4

Laptop-14:	Train	Validation	Test
Aspects	2041	256	634

Altered version used by Xin Li et al.

- CON class does not exist in dataset used by *Xin Li et al.*
- "re-prepared in *Li et al. (2019a)*" ([Xin Li et al.](#) page 4)

My replication

Original benchmark from: [Exploiting BERT for E2E-ABSA](#)

Model	LAPTOP		
	P	R	F1
BERT+Linear	62.16	58.90	60.43

Metrics from my replication

Replication:	precision	recall	f1
BERT+Linear	61.63	59.78	60.68

- My best replication is very close, slightly lower prediction and slightly higher recall
- Overall my replication had a slightly higher F1 score

Making Subsets of Laptop-14 Test

- Conflict subset was built using CLI carving for sentences with words but/however/though then hand cleaned fragments out.
 - This subset had 23 aspects
 - small sample due to limited number of sentences that qualified in dataset.
- Non-conflict subset was built by randomly selecting sentences by hand that did not include discourse or conflicting terms.
 - This subset had 21 aspects
 - small sample to attempt matching the subset size of pure-conflict subset.

Aggregated results

Test Type	Precision	Recall	micro-f1
*Full test set	61.63	59.78	60.68
non-conflict	78.95	71.43	74.99
conflict	48.15	56.52	51.99

- Model does not completely fail at predicting aspects and sentiment in pure-conflict sentences, though clearly hurts overall performance.
- 23 point spread from pure-conflict to non-conflict subsets

Error breakdown for conflict cases

Example 1: Model predictions vs. Actual Labels

Input → "The case is now slightly larger than the previous generation, but the lack of an external power supply justifies the small increase in size."

Output → [Aspect]:Polarity
['case']: NEG
['external', 'power', 'supply']: NEG
['size']: NEG

Truth → [Aspect]:Polarity
['case']:NEG
['external', 'power', 'supply']: NEU

Error breakdown for non-conflict cases

Example 2: Model predictions vs. Actual Labels

Input → "The battery life is excellent 6-7 hours without charging."

Output → [Aspect]:Polarity
['battery', 'life']: POS
['charging']: NEG

Truth → [Aspect]:Polarity
['battery', 'life']: POS

Conclusion:

- Aggregated metrics hid limitations on challenging sentences.
 - Pure-conflict reveals 8.69 drop
 - Non-conflict reveals 14.31 gain
 - Totalling a 23 point spread with the benchmark in the lower middle of the two extremes.
- Training material had few sentences with conflict or comparison structures, yet the model shows stability when tested on pure-conflict.
- Future work could explore these limitations further to determine which conflict types perform the worse and why.
- The dataset, Laptop-14, used as a benchmark for ABSA could be re-evaluated and updated with more robust rules for labeling to improve contextual extraction.